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Lab 3: Component Modelling & Architectural Pattern Selection

Project: Household Carbon Footprint & Target Tracker

Architecture Selection and Justification

Architecture Selection

We selected Layered Architecture for the Household Carbon Footprint & Target Tracker. The system separates user interaction, consumption management, carbon-emission calculation, target and progress management, and data storage into distinct components.

Reason 1 – Clear Separation of Responsibilities

Layered Architecture provides clear separation of concerns. The User Interface handles interaction with the resident, the Consumption Manager handles electricity and transport data, the Carbon Emission Calculator calculates CO₂e, the Target & Progress Manager manages reduction targets and milestones, and the Emission Data Repository stores and retrieves data. This makes the system easier to understand and maintain.

Reason 2 – Suitable for the System Workflow

The system follows a structured workflow in which monthly consumption data is entered, validated, processed to calculate CO₂e, stored, and then displayed to the user. Layered Architecture supports this organized flow between the different system components.

Security Advantage

Layered separation allows authorization and access-control checks to be applied before sensitive household consumption and carbon-footprint information is accessed. This helps prevent unauthorized users from accessing household data.

Performance Benefit

Separating data access, processing, and presentation allows historical emission data to be retrieved and processed efficiently before being displayed. This supports the requirement for responsive historical comparison and milestone charts.

Conclusion

Layered Architecture provides a simple and organized structure for the Household Carbon Footprint & Target Tracker. It offers clear separation of responsibilities, supports the system workflow, improves maintainability, and provides suitable security and performance characteristics for the project.